

Chapter 154 — Prostate Cancer Biomarkers — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Prostate-specific antigen (PSA)**: organ specific, **not cancer specific**.
- **5 α -reductase inhibitor (5ARI)** → PSA ↓ 50% at 12 months → double it.
- **Percentage free PSA (%fPSA)**: **low = cancer**; unchanged by 5ARI.
- **%fPSA approved range**: PSA 4–10, negative digital rectal examination (DRE).
- **PSA velocity** → never isolated biopsy trigger.
- **PSA density ≤ 0.15** → classic active surveillance (AS) criterion.
- **Prostate Health Index (PHI)** = $(p2PSA \div fPSA) \times \sqrt{tPSA}$.
- **4Kscore cut-off 7.5%**; below → low long-term metastasis risk.
- **Six National Comprehensive Cancer Network (NCCN) 2023 biomarkers**: IsoPSA, 4Kscore, PHI, SelectMDx, EPI, MPS.
- **BRCA2** → highest risk, younger, higher grade, worse survival.

INTRODUCTION

- **PSA discovered** 1979; PSA era → most cancers found at **T1c**.
- **ERSPC** (European Randomized Study of Screening for Prostate Cancer) and **Göteborg-1**: screening ↓ cancer-specific mortality; ↑ negative biopsies, overdiagnosis.
- **USPSTF 2012** (US Preventive Services Task Force): **Grade D** against screening.
- **Clinically significant** = Gleason score (GS) ≥ 7 = Grade Group (GG) ≥ 2 .
- **Generations**: 1st PSA (epithelium) → 2nd PCA3, fusions (cancer) → 3rd SchLAP1 (high grade).
- **First-line screening test**: PSA.
- **Guidelines**: secondary test (magnetic resonance imaging, MRI, or biomarker) before biopsy if PSA raised.
- **Three settings**: early detection; AS vs treatment; post-treatment.

BLOOD-BASED BIOMARKERS

- **PSA = human kallikrein 3 (hK3)**; family 15 serine proteases, **chromosome 19q13**.
- **Homology with PSA**: hK1 60%, hK2 ~78–80%.
- **Pre-pro-PSA** (17-aa leader) → **proPSA 244 aa** (7-aa pro) → **hK2** → active **PSA 237 aa**.
- **Function**: liquefies semen — cleaves semenogelin, fibronectin.
- **Source**: prostatic luminal epithelial cells; androgen dependent.
- **Half-life**: 2–3 days.
- **Tissue PSA peaks**: 0–6 months, >10 years; serum detectable from puberty.
- **PSA-A2M**: retains some protease activity.
- **Serum**: ~70% bound; free 5–45%.
- **PSA- α 1-antichymotrypsin (ACT)**: 60–90% of bound; inactive; detectable.
- **PSA- α 2-macroglobulin (A2M)**: 10–20%; **undetectable** (all epitopes masked).
- **PSA- α 1-protease inhibitor (API)**: 1–5%; detectable.
- **Elevation mechanism**: disrupted basal layer/basement membrane.
- **Most important cause of rise**: prostate disease (cancer, benign prostatic hyperplasia [BPH], prostatitis).
- **Primary screening confounders**: BPH, prostatitis.
- **After biopsy**: raised ≥ 4 weeks.
- **Cycling >55 km**: ~10% rise.
- **Ejaculation, age ≥ 50** : transient rise; baseline within 24 h → abstain, repeat.
- **Per cell**: cancer makes less PSA than benign.
- **Per cc serum PSA**: Gleason 3 $\approx$ benign; Gleason 4 several-fold higher.
- **Doubling rule**: overestimates first 6 months; underestimates after years.

- **PCPT** (Prostate Cancer Prevention Trial): 5ARI improves PSA screening performance.
- **PSA velocity**: normal 0.04 ng/mL/yr; BPH 0.07–0.27.
- **PSA density** = PSA $\div$ volume; biopsy cut-offs **0.08–0.15**.
- **Density after negative MRI**: 0.15 lacks specific supporting data.
- **Density in AS**: key predictor of reclassification.
- **%fPSA FDA indication**: total PSA 4–10, negative DRE.
- **NCCN**: %fPSA to improve specificity before repeat biopsy.
- **Risk calculators with fPSA**: PCPT, Prostate Biopsy Collaborative Group (PBCG).
- **%fPSA thresholds**: 15–25%; **25% → 95% sensitivity**.
- **%fPSA commonest use**: raised PSA after negative biopsy (21–28%).
- **%fPSA reports**: any-grade risk → overestimates GG ≥ 2 .
- **Free isoforms** (~equal): proPSA, benign PSA (BPSA), intact PSA (iPSA).
- **proPSA**: peripheral zone, cancer; **[–2]proPSA** most useful → PHI.
- **Truncated proPSAs**: [–2], [–4], [–5]; p2PSA cleaved Leu5–Ser6.
- **BPSA**: transition zone, BPH, volume; no prognostic value.
- **iPSA**: used within 4Kscore.
- **IsoPSA**: structure-based; K = bottom:top complexed PSA; unaffected by 5ARI, α -blockers.
- **4Kscore**: tPSA + fPSA + iPSA + **hK2** + age, prior biopsy, DRE.
- **hK2 vs PSA**: hK2 minimal in benign, intense in cancer; independent of PSA.
- **IsoPSA output**: KR = % risk GG ≥ 2 (0–100%).
- **4Kscore output**: % risk GG ≥ 2 ; adds ~0.11 area under curve (AUC) to clinical model.
- **4Kscore 6–15% thresholds**: 30–58% fewer biopsies.
- **4Kscore 7.5%**: sensitivity 94%, negative predictive value (NPV) 95%.
- **4Kscore origin**: ERSPC Göteborg (develop) → Rotterdam (validate).
- **PHI thresholds**: 24 or 28.6 proposed; **none validated**.
- **PHI 28.6**: 90% sensitivity GG ≥ 2 , ~30% biopsies avoided.
- **PHI reports**: any-grade, not GG ≥ 2 , risk.

URINE-BASED BIOMARKERS

- **PCA3** (prostate cancer antigen 3): long non-coding RNA, **chromosome 9q21–22**.
- **PCA3 score** = $(PCA3 \text{ mRNA} \div KLK3 \text{ mRNA}) \times 1000$; **post-DRE urine**.
- **PCA3 vs PSA**: **independent of prostate size**.
- **PCA3 thresholds**: 10, 25, 35; nomogram cut-off 17.
- **PCA3 threshold choice**: depends on prior biopsy status.
- **PCA3 evidence**: beats total PSA; no health-outcome benefit shown.
- **PCA3 FDA/CE indication**: prior negative biopsy.

- **PCA3 now:** not standalone → inside EPI and MPS.
- **Assay evolution:** reverse transcription polymerase chain reaction (RT-PCR) → transcription-mediated amplification (TMA).
- **SelectMDx:** DLX1 + HOXC6 mRNA, post-DRE, + clinical; **needs prostate volume.**
- **SelectMDx performance:** sensitivity ~93%, NPV ~95% GG ≥2.
- **ExoDx Prostate IntelliScore (EPI):** exosomal PCA3 + ERG + SPDEF; **no DRE, no clinical data.**
- **EPI thresholds: 15.6 initial; 20 repeat biopsy.**
- **EPI 15.6:** ~92% sensitivity; ~1/3 biopsies avoided; ~8% GG ≥2 missed.
- **EPI predecessor EXO106:** PCA3 + ERG; threshold 10.
- **MyProstateScore (MPS):** PCA3 + TMPRSS2:ERG + serum PSA; 0–100.
- **MPS thresholds: 10 initial** (97% sensitivity, 98% NPV); **15 repeat.**
- **MPS 10:** ~1/3 biopsies avoided; 3% GG ≥2 missed.
- **MPS validated:** Black men, suspicious DRE.
- **PCA3 + T2:ERG:** specificity 17% → 33% at 95% sensitivity.
- **TMPSRSS2:ERG (2005):** ~100% specific; ~50% of tumours, ~75% of men.
- **TMPSRSS2:** androgen regulated; partner ERG or ETV1 (ETS family).
- **miRNA:** 19–22 nucleotides; miR-141 (aggressive), urine miR-187.

TISSUE-BASED BIOMARKERS

- **Epigenetics:** DNA methylation, histone acetylation; CpG islands = GC-rich promoters.
- **Methylated genes:** GSTP1, APC, RARβ2, RASSF1A.
- **Field (halo) effect:** methylation beside tumour, not distant.
- **ConfirmMDx:** GSTP1, APC, RASSF1 → repeat-biopsy NPV ~88–90%.
- **Prolaris:** 31 cell cycle progression genes; predicts cancer-specific death.
- **Oncotype Dx Prostate:** 12 genes/4 pathways; adverse pathology = primary pattern 4 or pT3.
- **Decipher:** 22 markers; post-prostatectomy; ≥0.4 → **consider radiotherapy.**

- **Decipher risk groups:** low → intermediate → high 10-year metastasis.
- **Genomic classifier limit:** multifocality; data mostly retrospective.
- **PORTOS:** 24 genes; adjuvant radiotherapy response.
- **G-MAJOR:** prospective randomised trial, Decipher in favourable-risk.
- **Hereditary share:** up to 40–50%; familial 15–20% of cases.
- **One affected relative:** 2–4-fold risk.
- **>4 risk SNPs:** 4.5-fold; with family history 10-fold.
- **STHLM3:** 4 kallikreins + 232 SNPs + clinical.
- **BRCA2:** 4.5-fold <65; ~8.6-fold overall. **BRCA1:** 1.8-fold <65.
- **Lynch MSH2:** ~10-fold. **CHEK2:** OR ~2 (3.4 familial).
- **HOXB13 G84E:** 4.5-fold; **not** more aggressive.
- **Aggressive-disease genes:** BRCA1, BRCA2, ATM — DNA repair.
- **NCCN germline focus:** BRCA testing.
- **Multigene panel:** BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM.
- **ConfirmMDx, Black men:** NPV ~94% for GS ≥7.

CLASSIC TRAPS

- **Trap:** cancer cells make more PSA — **Actually:** less per cell; leak raises serum.
- **Trap:** low free PSA reassuring — **Actually:** low %fPSA signals cancer.
- **Trap:** 5ARI invalidates %fPSA — **Actually:** ratio unchanged.
- **Trap:** A2M-bound PSA measured — **Actually:** undetectable by most assays.
- **Trap:** rising velocity → biopsy — **Actually:** never an isolated trigger.
- **Trap:** doubling rule exact — **Actually:** overestimates early, underestimates late.
- **Trap:** one threshold fits all — **Actually:** repeat biopsy higher (EPI 20, MPS 15).
- **Trap:** BPSA marks cancer — **Actually:** BPH, transition zone.
- **Trap:** HOXB13 → aggressive cancer — **Actually:** risk only, not aggression.
- **Trap:** tissue classifier sees whole gland — **Actually:** sampled focus only.